

Home assignment: day 5 → day 8

Deadline: 2026-07-08, 16:59.

Note: you can always leave small number of binomial coefficients in the answer! Answer like $\binom{16}{11}$ is ok.

1. There are 20 chairs in a row and 7 students. Each student needs one chair to sit down. We do care only about which chairs are occupied.
 - a) In how many ways chairs can be occupied?
 - b) In how many ways chairs can be occupied if there should be at least one empty chair between occupied chairs?
 - c) In how many ways chairs can be occupied if there should be at least two empty chairs between occupied chairs?
2. A bridge hand (random 13 cards out of 52-card standard deck) is dealt.
 - a) What is the probability that the hand contains no spades?
 - b) What is the probability that the hand contains exactly 5 hearts?
 - c) What is the probability that the hand contains at most two aces?

Note: you can leave small number of binomial coefficients in the answer! Answer like $\binom{16}{11} / \binom{25}{13}$ is ok.

3. Solve the recurrence equation:

$$\begin{cases} a_n = 3a_{n-1} - 2b_{n-1} - 4, \\ b_n = -2a_{n-1} + 3b_{n-1} + 4, \\ a_0 = 1, \quad b_0 = 0. \end{cases}$$

4. I have a broken light switch button in my room. The switch button should turn the light on if the light is off and vice versa. On the first press the button works with probability $1/2$, on the second press — with probability $1/3$, on the third one — with probability $1/4$ and so on. Initially my room is dark.
 - (a) What is the probability that my room will be dark after the first press of button? After the second?
 - (b) What is the probability that my room will be dark after I press the button 100 times?

Demo for the quiz on day 6

1. Calculate the following complex values:

$$(3 + 2i) \cdot (2 - 5i), \quad \frac{3 - 7i}{3 - 4i}, \quad (1 + 3i)^3.$$

2. Consider the recurrence equation

$$a_n - 6a_{n-1} + 9a_{n-2} = 8n - 12.$$

Find the solution of the form $a_n = \alpha n + \beta$.